Guduru Sai Likhitha

ic24btech11007@iith.ac.in | likhitha09guduru@gmail.com | +91 7386539536 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.07
Narayana junior college	Class XII (Board of Intermediate Education,Andhra Pradesh)	2024	97.60%
Narayana School, KPHP Colony	Class X (Board of Secondary Education ,Telangana state)	2022	10.00%

SKILLS

Languages: Python, SQL, C++

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Git version control

Statistics & ML: Hypothesis testing, probability distributions, regression, sampling techniques -- Logistic Regression, Linear Regression, Decision Trees, Random Forest, Gradient Boosting (XGBoost/LightGBM), Support Vector Machines, k-Nearest Neighbors, Naive Bayes, K-Means Clustering, PCA, scikit-learn

Model Validation & Risk: AUC, Gini, KS, PSI, CSI, precision/recall, rank-order analysis, back-testing, out-of-fold cross-validation, champion-challenger benchmarking; assessing overfitting, data leakage, bias, and model drift

Deep Learning: PyTorch, PyTorch Geometric (GNNs), CNNs, TensorFlow, NumPy, Pandas

Backend & Tooling: Flask (REST APIs), Docker, GitHub Actions (CI), pytest, DVC, Git, Render

PROJECTS

Retail Credit Risk Scorecard -- Model Development, Validation & Champion-Challenger Benchmarking

Python, pandas, NumPy, scikit-learn, XGBoost

GitHub:
https://github.com/likhitha09guduru/Validation-of-Retail-Credit-Risk-Models-Logistic-Regression-vs.-Challenger-XGBoost

- Built an end-to-end retail credit scorecard pipeline (data prep, vintage sampling, segmentation, variable reduction, logistic regression model, score calibration, validation), mirroring a full retail ML model validation lifecycle

- Validated a Logistic Regression champion model against an XGBoost challenger using 5-fold out-of-fold cross-validation, benchmarking Gini, KS, PSI, and precision/recall on equal footing

- Assessed model discrimination and stability using Gini, KS, PSI, and Characteristic Stability Index (CSI) across 26 risk variables, with precision/recall/F1 evaluated at a matched approval-rate cutoff

- Recommended retaining the champion model based on validation evidence, documenting the challenger's overfitting risk under model governance standards

- Assessed data quality and sampling risk (missing values, outliers, feature stability, and reject inference) across merged bureau and customer datasets

Renal Tumor Classification -- Kidney CT Scan Image Classifier

Python, TensorFlow, Flask, DVC, Docker

GitHub: https://github.com/likhitha09guduru/Renal-Tumor-Classification-

- Built an end-to-end, config-driven ML pipeline (data ingestion , base model prep , training ,evaluation) enabling reproducible experimentation and easy hyperparameter iteration

- Structured the codebase into modular, reusable components (config, pipeline stages, entities) driven by YAML configuration files, separating logic from experiment parameters

- Fine-tuned a VGG16 CNN via transfer learning to classify kidney CT scans as Normal or Tumor, with configurable data augmentation, batch size, and learning rate

- Applied data augmentation (rotation, zoom, flips) to improve generalization on a limited medical imaging dataset and reduce overfitting

- Tracked pipeline stages, data/model artifacts, and evaluation metrics using DVC to ensure experiment reproducibility and version control across training runs

- Deployed the trained model via a Flask REST API with a web UI for real-time image upload and inference, containerized with Docker for portability

RELEVANT COURSES

Probability , Linear Algebra , Introduction to Programming , Calculus

WORK EXPERIENCE

Internship in Laboratory Chemical Safety at Indian Institute of Technology Hyderabad

- Developed a Laboratory Chemical Safety Manual covering safe chemical handling,waste disposal,and emergency response procedures.
- Researched and compiled laboratory safety guidelines and standard operating procedures(SOPs) for academic laboratory use.